

1 Introduction

We consider the following stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases} \quad (1.1)$$

where b is assumed to be a globally Lipschitz continuous function. $W(t, s)$ is a Gaussian noise that is white in time and colored in space. In general, when the initial measure satisfies the condition that

$$\int_{\mathbb{R}^d} \exp(-a|x|^2)|\mu|(dx) < \infty \quad \text{for any } a > 0 \quad (1.2)$$

and

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \text{for some and therefore any } \beta > 0 \quad (1.3)$$

then (1.1) has mild solution of the form

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(u(s, y))G(t - s, x - y)W(ds, dy) \quad (1.4)$$

The goal of this paper is to show the existence of an invariant measure for the solution (1.4) in some weighted $L^2_\rho(\mathbb{R}^d)$ space. [9] proved that under some integrability assumptions on the spectral density of the Gaussian noise W , the existence of an invariant measure for the solution (1.4) in two different scenarios:

1. The initial measure is a function, $\varphi \in L^2_\rho \cap L^2_{\hat{\rho}}$ where both ρ and $\hat{\rho}$ satisfy the condition that $\rho \cdot (\hat{\rho})^{-1} \in L^1(\mathbb{R}^d)$.
2. The initial measure is a constant function.

In this paper we replace the spectral density's integrability condition from [9] with a new condition allowing us to relax the conditions on the initial measure μ to include any $\varphi \in L^2_\rho$ and also the initial measure $\mu(\cdot) = \delta_0(\cdot)$.

This paper is broken into several sections. We first discuss some preliminary results and ideas needed to show the existence of an invariant measure. In the next section we state and prove several results regarding the existence of an invariant measure of the solution (1.4) inside the weighted L^2_ρ space. In this section we prove Theorem 3.1, which is the analog of Theorem 3.1 from [9]. We follow the blueprint laid out in [9] to prove this result, however we use the theory of Walsh and Dalang ([10] and [5]) to do so. The proof of our Theorem 3.1 relies on a technical lemma whose proof is very long and tedious. For this reason we postpone the proof of the lemma until the last section of this paper.

2 Preliminaries

Following [5], a Gaussian noise that is white in time and colored in space is an $L^2(\Omega, \mathcal{F}, P)$ -valued zero mean Gaussian Processes on a complete probability space $(\Omega, \mathcal{F}, P)$

$$\{F(\psi) : \psi \in C_0^\infty(\mathbb{R}^{d+1})\},$$

such that

$$\mathbb{E}[F(\psi)F(\varphi)] = J(\psi, \varphi) := \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\psi(s, \cdot) * \tilde{\varphi}(s, \cdot))(x)$$

where $\tilde{\varphi}(x) := \varphi(-x)$. When Γ has density $\Gamma(dx) = f(x)dx$ then J above can be written as

$$J(\psi, \phi) = \int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \psi(s, x) f(x - y) \varphi(s, y).$$

f is often referred to as the correlation function of the Gaussian Noise. Now following [10] and [5] and by denoting $\mathcal{B}_b(\mathbb{R}^d)$ as the family of bounded Borel subsets of $\mathbb{R}^d$, one may construct out of this a worthy martingale measure $W = (W_t(A), t \geq 0, A \in \mathcal{B}_b(\mathbb{R}^d))$ where

$$W_t(A) := \lim_{t \rightarrow \infty} F(\psi_n),$$

where the limit is in $L^2(\Omega, \mathcal{F}, P)$, $\psi_n \in C_0^\infty(\mathbb{R}^{d+1})$ and $\psi_n \downarrow 1_{[0,t] \times A}$. With this construction, one may show that $(W_t(A), t \geq 0, A \in \mathcal{B}_b(\mathbb{R}^d))$ is a worth martingale measure and its covariation measure Q is given by

$$Q(A \times B \times [s, t]) = (t - s) \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy 1_A(x) f(x - y) 1_B(y)$$

and its dominating measure is $K \equiv Q$. The main point is that F and W are related by

$$F(\psi) = \int_0^\infty \int_{\mathbb{R}^d} \psi(t, x) W(dt, dx)$$

where the integral is a martingale measure stochastic integral as in Walsh with filtration given by

$$\mathcal{F}_t = \sigma(W_s(A), s \leq t, A \in \mathcal{B}_b(\mathbb{R}^d)) \vee \mathcal{N}$$

where $\mathcal{N}$ is the sigma field generated by the P -null sets.

The arguments in this paper have been made possible due to the moment bounds of the solution (1.4) given in [3] which we will now state without proof. Below $\|\cdot\|_p$ denotes the standard $L^p(\Omega)$ norm. We will continue to use this notation throughout the paper.

Theorem 2.1. *Under conditions (1.2) and (1.3) and for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.1), $u(t, x)$, given by (1.4) is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq \text{const.} [1 + (|\mu| * G(t, \cdot))(x)] H(t; \gamma_p)^{1/2} \quad (2.1)$$

and moreover, when Assumption (i) stated below is assumed then we may make the stronger assertion that

$$\|u(t, x)\|_p \leq \text{const.} [1 + (|\mu| * G(t, \cdot))(x)] \quad (2.2)$$

Remark 2.1. *The definition of const. , γ_p and $H(t; \gamma_p)$ can be found in [3] however the actual meanings of these objects do not play a role in this paper. Throughout this paper we will in fact be working under Assumption (i) and so what will be important to us is (2.2). The proof of (2.1) can be found in [3] and the proof of (2.2) can be found in [4].*

We denote by $L_\rho^2(\mathbb{R}^d)$ the weighted space $L^2(\mathbb{R}^d, \sqrt{\rho(x)} dx)$ with norm

$$|f|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2 \rho(x) dx < \infty \quad (2.3)$$

and inner product

$$\langle f, g \rangle := \int_{\mathbb{R}^d} f(x) g(x) \rho(x) dx \quad (2.4)$$

For the remainder of the paper, we assume that ρ is an *admissible weight* meaning that ρ is a positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T) \rho(x) \quad \text{for all } t \in [0, T]. \quad (2.5)$$

We now state a series of additional assumptions that will need to be made throughout this paper.

Assumptions 2.1. (i)

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty$$

(ii) The function b satisfies the Lipschitz condition with Lipschitz constant Lip_b , namely

$$|b(u) - b(v)| < \text{Lip}_b |u - v| \quad u, v \in \mathbb{R}^d$$

(iii)

$$\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) < \infty$$

(iv)

$$\int_{\mathbb{R}^d} dx \rho(x) \left(\int_{\mathbb{R}^d} dy G(t_0, x-y) |\mu|(dy) \right)^2 < \infty \quad \text{for any } t_0 > 0$$

For the remainder of this section, we focus ourselves on (1.1) with initial condition $\varphi \in L^2_\rho$, namely the following stochastic heat equation.

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, x) = \varphi(x) & \varphi(x) \in L^2_\rho(\mathbb{R}^d) \end{cases} \quad (2.6)$$

which has mild solution given by

$$u(t, x) = \int_{\mathbb{R}^d} G_t(x-y) \varphi(y) dy + \int_0^t \int_{\mathbb{R}^d} b(u(s, y)) G(t-s, x-y) W(ds, dy) \quad (2.7)$$

We will now state the existence and uniqueness result for (2.6) which is proved in [3]. In addition, we will show that the solution, denoted as $u^\varphi(t, x)$, is an object in $L^2_\rho(\mathbb{R}^d)$ for each $t > 0$. We may denote the solution to (2.6) as $u(t, x)$ or simply as u when there is no confusion. First we state and prove a simple lemma showing that $\varphi \in L^2_\rho$ implies condition (1.2)

Lemma 2.1. Suppose that $\varphi \in L^2_\rho(\mathbb{R}^d)$. Then φ satisfies (1.2).

Proof: It can be shown that there exists a constant that for any admissible weight, there exists some constant $C > 0$ such that $C \cdot \rho(x) \geq \exp(-a|x|^2)$. This constant C may depend on $a > 0$. Now, since we assume $\int_{\mathbb{R}^d} \rho(x) |\varphi(x)|^2 dx < \infty$ then this implies that $\int_{\mathbb{R}^d} \rho(x) |\varphi(x)| dx < \infty$ which now implies that $\int_{\mathbb{R}^d} \exp(-a|x|^2) |\varphi(x)| dx < \infty$ $\square$

Theorem 2.2. Suppose that Assumption (i) and (ii) are satisfied. Then (2.7) gives the unique (in the sense of versions) random field solution $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$ of (2.6) starting from $\varphi(x)$. The solution is $L^2(\Omega)$ -continuous. Moreover, $u(t, \cdot) \in L^2_\rho(\mathbb{R}^d)$ almost surely for each $t > 0$.

Proof: Lemma 2.1 along with Assumptions (i) and (ii) guarantee the existence and uniqueness of the solution which is show in [3]. We only prove the assertion that $u(t, \cdot) \in L^2_\rho(\mathbb{R}^d)$ for each $t > 0$. Below we use C_i to represent a constant. Using the moment bounded (2.2) and the *admissible weight* property (2.5) we get the following.

$$\mathbb{E}(|u(t, \cdot)|^2_\rho) = \mathbb{E} \int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx$$

$$\begin{aligned}
&\leq C_1 \int_{\mathbb{R}^d} \left(1 + (|\varphi| * G(t, \cdot))(x) \right)^2 \rho(x) dx \quad \text{by the moment bounds (2.2)} \\
&\leq C_2 \int_{\mathbb{R}^d} \left(1 + (|\varphi| * G(t, \cdot))^2(x) \right) \rho(x) dx \\
&= C_2 \left[\int_{\mathbb{R}^d} \rho(x) dx + \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G_t(x-y) |\varphi(y)| dy \right)^2 \rho(x) dx \right] \\
&\leq C_3 \left[1 + \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G_t(x-y) |\varphi(y)| dy \right)^2 \rho(x) dx \right] \\
&\leq C_4 \left[1 + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G_t(x-y) |\varphi(y)|^2 \rho(x) dx dy \right] \\
&= C_4 \left[1 + \int_{\mathbb{R}^d} (\rho * G_t)(y) |\varphi(y)|^2 dy \right] \\
&\leq C_4 \cdot C(T) \left(1 + \int_{\mathbb{R}^d} \rho(y) |\varphi(y)|^2 dy \right) \quad \text{for any } T > 0, t \in [0, T] \text{ by the admissible weight property (2.5)} \\
&< \infty \quad \text{for any } T > 0, t \in [0, T]
\end{aligned}$$

Hence $u(t, \cdot) \in L_\rho^2(\mathbb{R}^d)$ almost surely for any given $t > 0$. $\square$

We finish off this section with a proposition on the functions $G(t, x)$ which will be used later on. The proof of this result can be found in [9]

Proposition 1. *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi(x) \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 – semigroup on L_ρ^2 . Moreover, if $\hat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty$$

then for any $t < 0$, the operators defined above are compact from $L_{\hat{\rho}}^2(\mathbb{R}^d)$ to $L_\rho^2(\mathbb{R}^d)$.

3 Existence of Invariant Measures

Let $(E, \mathcal{E})$ denote an arbitrary measure space and $\mathcal{M}_1(E)$ be the set of all probability measure on $(E, \mathcal{E})$. Let $P_t(x, A)$, $t \geq 0$, $x \in E$, $A \in \mathcal{E}$ be a Markovian Transition function as in [7]. We say that a measure $\eta \in \mathcal{M}_1(E)$ is invariant for P_t if

$$\eta(A) = \int_E P_t(x, A) \eta(dx) \quad \text{for each } t \geq 0.$$

Following [7], we define Markovian transition functions $P_t(\varphi, A) := \mathcal{L}(u^\varphi(t, \cdot))(A)$ for $\varphi \in L_\rho^2(\mathbb{R}^d)$, $t \geq 0$ and $A \in \mathcal{B}(L_\rho^2(\mathbb{R}^d))$ where $\mathcal{B}(L_\rho^2(\mathbb{R}^d))$ are the Borel subsets of the space $L_\rho^2(\mathbb{R}^d)$ and $\mathcal{L}(u^\varphi(t, \cdot))(A)$ denotes the laws of the solution of (2.6)

$$\mathcal{L}(u^\varphi(t, \cdot))(A) = P\{\omega \in \Omega : u^\varphi(t, \cdot) \in A \mid A \in \mathcal{B}(L_\rho^2(\mathbb{R}^d))\}$$

which define a probability measure on $\mathcal{B}(L_\rho^2(\mathbb{R}^d))$. For the case when the initial condition is a measure, as in (1.1) the laws are defined for $t > 0$ where t needs to be strictly greater than 0. This is indeed true by Theorem 3.2 below which states that under certain assumptions, the solution $u(t, \cdot)$ of (1.1) is almost surely in L_ρ^2 for every $t > 0$.

We start by stating a theorem guaranteeing the existence of an invariant measure as well as outlining the proof. The proof, which we reproduce using [9], relies on a series of lemmas which will be proven as well.

Theorem 3.1. *Assume that Assumptions (i) - (iii) hold true and that for an admissible weight ρ , there exists another admissible weight $\hat{\rho}$ and an element $\hat{\varphi} \in L^2_\rho(\mathbb{R}^d) \cap L^2_{\hat{\rho}}(\mathbb{R}^d)$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty \quad \text{and } u^{\hat{\varphi}}(\cdot, x) \text{ is bounded in probability in } L^2_{\hat{\rho}}(\mathbb{R}^d).$$

Then for any $\delta_0 > 0$ there exists an invariant measure for the laws of (2.7), $\{\mathcal{L}(u^{\hat{\varphi}}(t, \cdot))\}_{t \geq \delta_0}$, in $L^2_\rho(\mathbb{R}^d)$

Proof: We start by stating a lemma that will be used to complete the proof. The proof of the lemma is very technical and serves no real purpose in understanding its use to complete the proof of Theorem 3.1. For that reason we postpone proving it until later.

Lemma 3.1. *For arbitrary $\epsilon \in (0, 1)$, $\delta_0 > 0$ and $R > 0$ there exists a compact set $\mathcal{K}$ in L^2_ρ such that*

$$\mathbf{P}\{u^{\hat{\varphi}}(t, \cdot) \in \mathcal{K}\} \geq (1 - \epsilon) \mathbf{P}\{|u^{\hat{\varphi}}(t - \delta_0, \cdot)|_{\hat{\rho}} < R\} \quad \text{for all } t > \delta_0 \quad (3.1)$$

We may now complete the proof of Theorem 3.1. By Lemma 3.1, given an $\epsilon \in (0, 1)$ and $R > 0$ then we could find a compact set $\mathcal{K} \in L^2_\rho$ such that

$$\mathbf{P}\{u^{\hat{\varphi}}(t, \cdot) \in \mathcal{K}\} \geq (1 - \epsilon) \mathbf{P}\{|u^{\hat{\varphi}}(t - \delta_0, \cdot)|_{\hat{\rho}} < R\} \quad \text{for all } t > \delta_0$$

The boundedness in probability of $u^{\hat{\varphi}}(\cdot, x)$ then implies that the family of laws $\mathcal{L}(u^{\hat{\varphi}}(t, \cdot))$ for $t \geq \delta_0$ is tight in L^2_ρ and therefore the family of probability measures

$$\frac{1}{T} \int_{\delta_0}^{T+\delta_0} \mathcal{L}(u^{\hat{\varphi}}(t, \cdot)) dt, \quad T > 0 \quad (3.2)$$

is tight as well. Referring to **Corollary 3.1.2** from [7] we may say that (3.2) has a subsequence that weakly converges to some measure η and that in fact η is invariant for the laws of (2.7). $\square$

We end this section with another theorem guaranteeing the existence of an invariant measure for the solution to (1.1) with more general initial conditions.

Theorem 3.2. *Consider the stochastic heat equation (1.1) and suppose that assumptions (i) - (iv) hold true. Then for any $t_0 > 0$ $u(t_0, \cdot) \in L^2_\rho$ almost surely for any admissible weight. Also the solution to the following stochastic heat equation*

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta_x v(t, x) = b(x, v(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ v(0, x) = u(t_0, x) \end{cases} \quad (3.3)$$

satisfies the property that

$$\sup_{t \geq 0} \mathbb{E}|v(t, \cdot)|_\rho^2 < \infty \quad (3.4)$$

and in particular, (3.4) implies that v is bounded in probability in L^2_ρ .

Proof: We first show that $u(t_0, x) \in L^2_\rho$ for any admissible weight.

$$\begin{aligned} \mathbb{E}|u(t_0, \cdot)|_\rho^2 &= \int_{\mathbb{R}^d} dx \|u(t_0, x)\|_2^2 \rho(x) \\ &\leq k \int_{\mathbb{R}^d} dx (1 + (|\mu| * G(t_0, \cdot)(x)))^2 \rho(x) \quad \text{moment estimates} \\ &= k \int_{\mathbb{R}^d} dx \left(1 + \int_{\mathbb{R}^d} |\mu|(dy) G(t_0, x - y)\right)^2 \rho(x) \end{aligned}$$

$$\begin{aligned} &\leq k \left(\int_{\mathbb{R}^d} \rho(x) \, dx + \int_{\mathbb{R}^d} dx \, \rho(x) \left(\int_{\mathbb{R}^d} |\mu|(dy) G(t_0, x - y) \right)^2 \right) \\ &< \infty \quad \text{by assumption (iv)} \end{aligned}$$

This shows that $u(t_0, x) \in L_\rho^2$ almost surely for any admissible weight. Now it remains to show (3.4).

$$\begin{aligned} \mathbb{E}|v(t, \cdot)|_\rho^2 &= \int_{\mathbb{R}^d} \|v(t, x)\|_2^2 \rho(x) \, dx \\ &\leq k \int_{\mathbb{R}^d} (1 + (\|u(t_0, \cdot)\|_2 * G(t, \cdot)(x)))^2 \rho(x) \, dx \\ &= k \int_{\mathbb{R}^d} \left(1 + \int_{\mathbb{R}^d} \|u(t_0, y)\|_2 G(t, x - y) \, dy \right)^2 \rho(x) \, dx \\ &\leq k \int_{\mathbb{R}^d} \left[1 + \int_{\mathbb{R}^d} \left(1 + (|\mu| * G(t_0, \cdot)(y)) \right) G(t, x - y) \, dy \right]^2 \rho(x) \, dx \\ &= k \int_{\mathbb{R}^d} \left[1 + \int_{\mathbb{R}^d} \left(1 + \int_{\mathbb{R}^d} G(t_0, y - z) |\mu|(dz) \right) G(t, x - y) \, dy \right]^2 \rho(x) \, dx \\ &= k \int_{\mathbb{R}^d} \left[1 + \int_{\mathbb{R}^d} G(t, x - y) \, dy + \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(t_0, y - z) G(t, x - y) \, dy |\mu|(dz) \right]^2 \rho(x) \, dx \\ &= k \int_{\mathbb{R}^d} \left(2 + \int_{\mathbb{R}^d} G(t + t_0, x - z) |\mu|(dz) \right)^2 \rho(x) \, dx \\ &\leq k \int_{\mathbb{R}^d} 4\rho(x) \, dx + \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t + t_0, x - z) |\mu|(dz) \right)^2 \rho(x) \, dx \\ &< \infty \quad \text{by assumption (iv) and } t \mapsto G(t_0 + t, x) \text{ is bounded by } G(t_0, x) \text{ up to a constant} \end{aligned}$$

This completes the proof of (3.4). To finish the proof we now show that (3.4) implies that $v(\cdot, x)$ is bounded in probability in L_ρ^2 . First let $\sup_{t \geq 0} \mathbb{E}|v(t, \cdot)|_\rho = B$. Let $\epsilon > 0$ then we may choose $R > 0$ large enough so that

$$\mathbf{P}[|v(t, \cdot)|_\rho \geq R] \leq R^{-2} \mathbb{E}|v(t, \cdot)|_\rho^2 \leq R^{-2} B < \epsilon$$

Hence $v(t, \cdot)$ is bounded in probability in L_ρ^2 . □

Corollary 3.1. *In addition to assumptions (i) - (iv), suppose there existed another admissible weight $\hat{\rho}$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} \, dx < \infty$$

and $u(t_0, x) \in L_{\hat{\rho}}^2(\mathbb{R}^d)$ where $u(t_0, x)$ is the same as in Theorem 3.2. Then there exists an invariant measure for the laws of the solutions to (1.1) for $t > \delta > 0$. In particular, for any $\delta > 0$ there exists a measure that is invariant for $\{\mathcal{L}(u^\mu(t, \cdot))\}_{t > \delta}$ where u is the solution to (1.1) with initial measure μ .

Proof: Pick $t_0 > 0$ and $\delta_0 > 0$ such that $\delta = t_0 + \delta_0$ for a given $\delta > 0$. By Theorem 3.1, there exists an invariant measure, η , for $\{\mathcal{L}(v(t, \cdot))\}_{t \geq \delta_0}$ where v is the solution to (3.3). However, by the semi group property of the heat solution, $v(t, x) = u^\mu(t + t_0, x)$ and so η is invariant for $\{\mathcal{L}(u^\mu(t + t_0, \cdot))\}_{t \geq \delta_0} = \{\mathcal{L}(u^\mu(t, \cdot))\}_{t \geq \delta}$. □

Corollary 3.2. *Under assumptions (i) - (iii) and for any $\delta > 0$, the stochastic heat equation (2.6) and also the following stochastic heat equation*

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \delta_0(\cdot) \end{cases} \quad (3.5)$$

have an invariant measure in L_ρ^2 , $\rho(x) = \exp(-2|x|)$, for the laws of their solutions for $t > \delta$

Proof: Both SPDEs above satisfy assumption (iv). We consider the weights $\rho(x) = \exp(-2|x|)$ and $\hat{\rho}(x) = \exp(-|x|)$ which clearly satisfy

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty.$$

By Theorem 3.2 we know that for an $t_0 > 0$, the solution at $t = t_0$ from the two SPDE's in question are in $L_\rho^2 \cap L_{\hat{\rho}}^2$. Corollary 3.1 implies the result. $\square$

Remark 3.1. In theorem 3.1 we showed the existence of an invariant measure for the laws of the solution of (2.6) for $t > 1$. However in that theorem we needed to assume a bounded in probability condition. Now in Corollary 3.2, we prove that there exists an invariant measure for the solution of (2.6) for $t \geq \delta > 0$ where we now can drop the bounded in probability assumption in exchange for (iv). So for every $\varphi \in L_\rho^2$, the laws of the solution $u^\varphi(t, x)$ have an invariant measure.

4 Proof of Lemma 3.1

We first define two operators. Let $q > 2$ and $t_0 > 0$ be arbitrary and $\alpha \in (q^{-1}, 2^{-1})$. Define the operator $F_\alpha^{t_0}$ as follows

$$(F_\alpha^{t_0} f)(x) := \int_0^{t_0} ds \int_{\mathbb{R}^d} dy (t_0 - s)^{\alpha-1} G(t_0 - s, x - y) f(s, y) \quad (4.1)$$

It is shown in [9] that F_α is compact from $L^q((0, t_0), L_\rho^2)$ to L_ρ^2 where $L^q((0, t_0), L_\rho^2)$ is the space of functions that map $s \mapsto f(s, \cdot)$ and $f(s, \cdot) \in L_\rho^2$ and in addition $\int_0^{t_0} |f(s, \cdot)|_\rho^q ds < \infty$.

Next we define for $\varphi \in L_\rho^2$ the operators $Y^\varphi(s, y)$, or simply as $Y(s, y)$ when there is no confusion, as follows

$$Y^\varphi(s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(u^\varphi(r, z)) W(dr, dz) \quad (4.2)$$

Lemma 4.1. Let $q > 2$, then Y^φ operators, (4.2), satisfies the inequality

$$\mathbb{E} \int_0^1 |Y^\varphi(s, y)|_\rho^q ds \leq K |\varphi|_\rho^q \quad (4.3)$$

Proof: First, notice by applying Minkowski's Integral Inequality that

$$\begin{aligned} E|Y(s, y)|_\rho^q &= \int_\Omega dP \left(\int_{\mathbb{R}^d} dy Y(s, y)^2 \rho(y) \right)^{q/2} \\ &\leq \left(\int_{\mathbb{R}^d} dy \left(\int_\Omega dP Y(s, y)^q \rho(y)^{q/2} \right)^{2/q} \right)^{q/2} \\ &= \left(\int_{\mathbb{R}^d} dy \rho(y) \|Y(s, y)\|_q^2 \right)^{q/2} \\ &= \left\| \|Y(s, y)\|_q \right\|_\rho^q \end{aligned}$$

Because of this, we look to find an appropriate upper bound for $\|Y(s, y)\|_q^2$ which for now we denote as $M(s, y)$. Then we integrate this bound

$$\int_{\mathbb{R}^d} M(s, y) \rho(y) dy$$

which gives us a bound for $\left| \|Y(s, y)\|_q \right|_\rho$. To do this, we start by applying the Burkholder-Davis-Gundy Inequality from [5] as well as Minkowski's Integral Inequality to see that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C_q \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(u(r, z_1))\|_q f(z_1 - z_2) \\ &\quad \times G(s-r, y-z_2) \|b(u(r, z_2))\|_q \end{aligned}$$

Note that for Lipschitz functions, we have that $|b(u)| < k(1 + |u|)$. We apply this and the moment bound (2.2) to $\|b(u(r, z_i))\|_q$ above to see that

$$\begin{aligned} \|b(u(r, z_i))\|_q &\leq k(1 + \|u(r, z_i)\|_q) \\ &\leq k_1 \left(1 + (1 + J_0(r, z_i)) \right) \quad \text{by (2.2)} \\ &\leq k_2(1 + J_0(r, z_i)) \end{aligned}$$

Now applying this to the inequality for $\|Y^\alpha(s, y)\|_q^2$ above we get that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq k_3 \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) (1 + J_0(r, z_1)) f(z_1 - z_2) \\ &\quad \times G(s-r, y-z_2) (1 + J_0(r, z_2)) \\ &= \underbrace{k_3 \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2)}_{I_1} \\ &\quad + \underbrace{k_3 \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_1)}_{I_2} \\ &\quad + \underbrace{k_3 \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_2)}_{I_3} \\ &\quad + \underbrace{k_3 \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_1) J_0(r, z_2)}_{I_4} \end{aligned}$$

We now investigate each of the integrals I_k above. Throughout the remainder of this proof and the proof of Lemma 4.2 we will be using the following relation

$$G(s-r, y-z_i) G(r, z_i - \sigma) = G(s, y - \sigma) G\left(\frac{r(s-r)}{s}, z_i - \sigma - \frac{r}{s}(y - \sigma)\right) \quad (4.4)$$

For I_1 :

$$\begin{aligned} I_1 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \Gamma(d\xi) (G(s-r, y - \cdot) * G(s-r, y + \cdot))(\xi) \\ &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \end{aligned}$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we get that

$$\begin{aligned} \int_{\mathbb{R}^d} I_1 \cdot \rho(y)dy &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \int_{\mathbb{R}^d} \rho(y)dy \\ &\leq (2\pi)^{-2d} \cdot \|\rho\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \end{aligned}$$

For I_2 :

$$\begin{aligned} I_2 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) J_0(r, z_1) \\ &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) \int_{\mathbb{R}^d} d\sigma G_r(z_1-\sigma) |\varphi(\sigma)| \\ &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_2) f(z_1-z_2) G\left(\frac{r(s-r)}{s}, z_1-\sigma-\frac{r}{s}(y-\sigma)\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\ &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \end{aligned}$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we get that

$$\begin{aligned} \int_{\mathbb{R}^d} I_2 \cdot \rho(y)dy &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} dy \rho(y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\ &\leq (2\pi)^{-2d} \cdot C(T) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma \rho(\sigma) |\varphi(\sigma)| \\ &= (2\pi)^{-2d} \cdot C(T) \cdot \|\varphi\|_{L^1_\rho(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \end{aligned}$$

For I_3 : By completing the same steps as for I_2 we get that

$$I_3 \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|$$

and now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we will get the same bound as in for I_2 above. Namely,

$$\int_{\mathbb{R}^d} I_3 \cdot \rho(y)dy \leq (2\pi)^{-2d} \cdot C(T) \cdot \|\varphi\|_{L^1_\rho(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right)$$

For I_4 :

$$\begin{aligned} I_4 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) J_0(r, z_1) J_0(r, z_2) \\ &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) \\ &\quad \times \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(r, z_1-\sigma_1) G(r, z_2-\sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)| \\ &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(s, y-\sigma_1) G(s, y-\sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)| \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1-z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i-\sigma_i-\frac{r}{s}(y-\sigma_i)\right) \\ &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(s, y-\sigma_1) G(s, y-\sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)| \end{aligned}$$

$$\begin{aligned}
& \times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \\
& = (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \left(\int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|\right)^2 \\
& \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|^2
\end{aligned}$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we get that

$$\begin{aligned}
\int_{\mathbb{R}^d} I_4 \cdot \rho(y) dy & \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} dy \rho(y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|^2 \\
& \leq (2\pi)^{-2d} \cdot C(T) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma \rho(\sigma) |\varphi(\sigma)|^2 \\
& = (2\pi)^{-2d} \cdot C(T) \cdot |\varphi|_\rho^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right)
\end{aligned}$$

Now, putting all of this together, we see that

$$\begin{aligned}
(E|Y(s, y)|_\rho^q)^{2/q} & \leq \int_{\mathbb{R}^d} dy \rho(y) \|Y(s, y)\|_q^2 \\
& \leq \int_{\mathbb{R}^d} dy \rho(y) (I_1 + I_2 + I_3 + I_4) \\
& \leq (2\pi)^{-2d} \cdot \|\rho\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\
& \quad + 2(2\pi)^{-2d} \cdot C(T) \cdot \|\varphi\|_{L_\rho^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)|\xi|^2\right) \\
& \quad + (2\pi)^{-2d} \cdot C(T) \cdot |\varphi|_\rho^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right)
\end{aligned}$$

and by assumption (iii) and Lemma 4.2 below, the three integrals above are all finite. Therefore there exists some constant K_T dependent on T such that

$$E|Y(s, y)|_\rho^q \leq K_T \cdot |\varphi|_\rho^q$$

Lastly if we integrate out s from $0 \rightarrow 1$ we get our desired inequality,

$$\int_0^1 E|Y(s, y)|_\rho^q ds \leq \int_0^1 K_T \cdot |\varphi|_\rho^q ds = K_T \cdot |\varphi|_\rho^q$$

□

Lemma 4.2. *Assuming (iii) then we may say the following two integrals are finite*

$$\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \quad (4.5)$$

and

$$\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left[-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)|\xi|^2\right] \quad (4.6)$$

Proof: We first focus on (4.5). By applying the substitution $u = s - r$ the we may rewrite

$$(iii) = \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-u|\xi|^2)$$

and

$$(4.5) = \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{u(s-u)}{s}|\xi|^2\right)$$

Since $u(s-u) \geq \frac{us}{2}$ for all $u \in [0, s]$ then we may say that

$$\begin{aligned} (4.5) &\leq \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{u}{2}|\xi|^2\right) \\ &= 2^{1-2\alpha} \int_0^{s/2} dt t^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-t|\xi|^2) \quad \text{substitution } t = \frac{u}{2} \\ &\leq (iii) < \infty \end{aligned}$$

which completes the proof showing assumption (iii) implies that (4.5) is finite. The finiteness of (4.6) follows easily by noting that

$$\begin{aligned} (4.6) &\leq \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{s-r}{2}|\xi|^2\right) \\ &= 2^{1-2\alpha} \int_0^{s/2} du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-u|\xi|^2) \quad \text{using the substitution } u = \frac{s-r}{2} \\ &\leq (iii) < \infty \end{aligned}$$

which completes the proof showing that assumption (iii) implies (4.6). $\square$

Lemma 4.3. *The following factorization holds for $\alpha \in (0, 1)$*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz) \quad (4.7)$$

Proof:

$$\begin{aligned} &\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds \\ &\stackrel{1}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(u(r, z)) W(dr, dz) \\ &\stackrel{2}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) W(dr, dz) \\ &\stackrel{3}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &\stackrel{4}{=} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz) \end{aligned}$$

where the last step is true provided $\alpha \in (0, 1)$. $\square$

Remark 4.1. *We also need to verify the use of Fubini's Theorem which is stated in [10]. We used Fubini's Theorem twice above: first from equality 1 to 2 and then from equality 3 to 4. For reference, we will state Fubini's Theorem as given in [10] and verify the two uses of it in the proof of Lemma 4.2.*

Theorem 4.1 (Fubini's Theorem). *Let $(G, \mathcal{G}, \mu)$ be a finite measure space and $(E, \mathcal{E}, \nu)$ be a σ -finite measure space. Let $g(z, s, \omega, y)$, $z \in E$, $s \geq 0$, $\omega \in \Omega$, $y \in G$ be a $\mathcal{E} \times \mathcal{G}$ measurable function. Suppose that*

$$\mathbb{E} \left[\int_{E \times E \times [0, T] \times G} |g(z_1, r, \omega, y) g(z_2, r, \omega, y)| K(dz_1, dz_2, dr) \mu(dy) \right] < \infty \quad (4.8)$$

where K is the dominating measure as in [5].

Remark 4.2. Following [5], the dominating measure for this problem is given by

$$K(dz_1, dz_2, dr) = f(z_1 - z_2) dz_1 dz_2 dr$$

where f is the correlation function of the spatially homogeneous Gaussian noise.

We now verify (4.8) for both applications of Fubini's Theorem.

From Equality 1 to 2:

Proof: We let

$$\begin{aligned} g(z, r, \omega, y) &= (s - r)^{-\alpha} G(s - r, y - z) b(u(r, z)) \\ (G, \mathcal{G}, \mu) &= (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), G_{t-s}(x - y) dy) \\ (E, \mathcal{E}, \nu) &= (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda) \end{aligned}$$

and so (4.8) becomes

$$\begin{aligned} (4.8) &= \mathbb{E} \int_{\mathbb{R}^d} dy G_{t-s}(x - y) \int_0^s dr (s - r)^{-2\alpha} \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) |b(u(r, z_1))| f(z_1 - z_2) G(s - r, y - z_2) |b(u(r, z_2))| \end{aligned}$$

For simplicity, we first investigate the $\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$ part of the integral. We continue by applying the Lipschitz property of b .

$$\begin{aligned} &\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) |b(u(r, z_1))| f(z_1 - z_2) G(s - r, y - z_2) |b(u(r, z_2))| \\ &\leq k_1 \cdot \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) (1 + u(r, z_1)) f(z_1 - z_2) G(s - r, y - z_2) (1 + u(r, z_2)) \\ &\leq k_1 \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) f(z_1 - z_2) G(s - r, y - z_2)}_{L_1} \\ &\quad + k_1 \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) \|u(r, z_1)\|_1 f(z_1 - z_2) G(s - r, y - z_2)}_{L_2} \\ &\quad + k_1 \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) f(z_1 - z_2) G(s - r, y - z_2) \|u(r, z_2)\|_1}_{L_3} \\ &\quad + k_1 \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) \|u(r, z_1)\|_2 f(z_1 - z_2) G(s - r, y - z_2) \|u(r, z_2)\|_2}_{L_4} \end{aligned}$$

For L_1 : If we back at I_1 from the proof of Corollary 4.1 then we see that

$$L_1 \leq (2\pi)^{-2d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s - r)|\xi|^2)$$

And now if we include the $\int_{\mathbb{R}^d} dy G_{t-s}(x - y) \int_0^s dr (s - r)^{-2\alpha}$ part of the integral we get that

$$\int_{\mathbb{R}^d} dy G_{t-s}(x - y) \int_0^s dr (s - r)^{-2\alpha} L_1 \leq (2\pi)^{-2d} \int_0^s dr (s - r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s - r)|\xi|^2)$$

For L_2 : We first use the moment bounds from [3] along with the fact that $\|u(r, z_1)\|_1 \leq \|u(r, z_1)\|_2$

$$\begin{aligned} L_2 &\leq k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)(1+J_0(r, z_1))f(z_1-z_2)G(s-r, y-z_2) \\ &= kL_1 + \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2) \end{aligned}$$

But this last integral above is the same thing as I_2 from Corollary 4.1 so we see that

$$L_2 \leq kL_1 + (2\pi)^{-2d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)|$$

And now if we include the $\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha}$ part of the integral we get that

$$\begin{aligned} &\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} L_2 \\ &\leq k \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} L_1 \\ &\quad + (2\pi)^{-2d} \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)| \\ &\leq k \cdot (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ &\quad + (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)| \\ &= k \cdot (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ &\quad + (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \underbrace{\int_{\mathbb{R}^d} d\sigma G(t, x-\sigma)|\varphi(\sigma)|}_{\text{continuous function of } x \text{ and } t} \\ &\leq k \cdot (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ &\quad + (2\pi)^{-2d} \cdot k_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \end{aligned}$$

For L_3 : Following the same steps for L_2 will get you the same exact bound as for L_3 .

For L_4 : We continue as we did for L_2 to see that

$$\begin{aligned} L_4 &\leq k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)(1+J_0(r, z_1))f(z_1-z_2)G(s-r, y-z_2)(1+J_0(r, z_2)) \\ &= k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)f(z_1-z_2)G(s-r, y-z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)f(z_1-z_2)G(s-r, y-z_2)J_0(r, z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2)J_0(r, z_2) \end{aligned}$$

These first three integrals have already been handled above in L_1, L_2 and L_3 of this proof so we only need to proceed with the fourth integral. However, notice from I_4 above in the proof of Corollary 4.1 that this

fourth integral is bounded by the following

$$\begin{aligned} & \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) J_0(r, z_1) f(z_1-z_2) G(s-r, y-z_2) J_0(r, z_2) \\ & \leq (2\pi)^{-2d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|^2 \end{aligned}$$

And now if we include the $\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha}$ part of the integral we get that

$$\begin{aligned} & \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|^2 \\ & = \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|^2 \\ & = \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \underbrace{\int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)|^2}_{\text{continuous function of } x \text{ and } t} \\ & \leq k_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \end{aligned}$$

And now putting this all together for L_4 we see that

$$\begin{aligned} \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} L_4 & \leq k \cdot (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ & \quad + 2 \cdot k \cdot (2\pi)^{-2d} \cdot k_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \\ & \quad + (2\pi)^{-2d} \cdot k \cdot k_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \end{aligned}$$

Considering the above arguments we may say that there is a constant $K_{x,t}$ depending on x and t such that (4.8) has the following bound which is finite by assumption (iii)

$$\begin{aligned} (4.8) & \leq K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ & \quad + K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \\ & \quad + K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \end{aligned}$$

and by referring to Lemma 4.2, this completes the justification for the first use of Fubini's Theorem.

From Equality 2 to 3:

Proof: We let

$$\begin{aligned} g(z, r, \omega, y) & = (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) \\ (G, \mathcal{G}, \mu) & = ((0, t), \mathcal{B}(0, t), (t-s)^{\alpha-1} ds) \\ (E, \mathcal{E}, \nu) & = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda) \end{aligned}$$

and so (4.8) become

$$(4.8) = \mathbb{E} \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) G(t-r, x-z_2) f(z_1-z_2) |b(u(r, z_1))| |b(u(r, z_2))|$$

and again for simplicity we will first investigate the $\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$ integral. Using the Lipschitz property, we similarly get that

$$\begin{aligned}
& \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) |b(u(r, z_1)) f(z_1-z_2) G(t-r, x-z_2) |b(u(r, z_2))| \\
& \leq k \cdot \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) (1+u(r, z_1)) f(z_1-z_2) G(t-r, x-z_2) (1+u(r, z_2)) \\
& \leq k \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) f(z_1-z_2) G(t-r, x-z_2)}_{K_1} \\
& \quad + k \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) \|u(r, z_1)\|_1 f(z_1-z_2) G(t-r, x-z_2)}_{K_2} \\
& \quad + k \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) f(z_1-z_2) G(t-r, x-z_2) \|u(r, z_2)\|_1}_{K_3} \\
& \quad + k \cdot \underbrace{\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) \|u(r, z_1)\|_2 f(z_1-z_2) G(t-r, x-z_2) \|u(r, z_2)\|_2}_{K_4}
\end{aligned}$$

We apply similar techniques as in for the first justification of Fubini's Theorem to get that

For K_1 :

$$\begin{aligned}
& \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_1 \leq \int_0^t ds (t-s)^{\alpha-1} \underbrace{\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(t-r)|\xi|^2)}_{\text{continuous function of } s \in [0, t] \text{ by Lemma 4.2}} \\
& \leq k_t \int_0^t ds (t-s)^{\alpha-1}
\end{aligned}$$

For K_2 and K_3 :

$$\begin{aligned}
& \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_2 \\
& \leq k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(t-r)|\xi|^2) \\
& \quad + k \int_0^t ds (t-s)^{\alpha-1} \underbrace{\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{t-r}{2} + \frac{r(t-r)}{2t}\right)|\xi|^2\right)}_{\text{continuous function of } s \text{ by Lemma 4.2}} \underbrace{\int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)|}_{J_0(t, x)}
\end{aligned}$$

The first integral is the same as in K_1 above and the second integral is bounded by

$$J_0(t, x) \cdot k_t \int_0^t ds (t-s)^{\alpha-1}$$

and hence the bound for K_2 and K_3 is

$$\int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_2 \leq k_t \int_0^t ds (t-s)^{\alpha-1} + J_0(t, x) \cdot k_t \int_0^t ds (t-s)^{\alpha-1}$$

For K_4 :

$$\begin{aligned}
& \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_4 \\
& \leq k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(t-r)|\xi|^2) \\
& \quad + 2k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{t-r}{2} + \frac{r(t-r)}{2t}\right)|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)| \\
& \quad + k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t}|\xi|^2\right) \left(\int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)|\right)^2 \\
& \leq k_t \int_0^t ds (t-s)^{\alpha-1} + 2k_t J_0(t, x) \int_0^t ds (t-s)^{\alpha-1} + k_t J_0(t, x)^2 \int_0^t ds (t-s)^{\alpha-1}
\end{aligned}$$

Now we may say that that there exists some constant K_t dependent on t such that (4.8) has the following bound which is finite since $0 < \alpha < 1$

$$(4.8) \leq K_t \left(1 + 2J_0(t, x) + J_0(t, x)^2\right) \int_0^t ds (t-s)^{\alpha-1}$$

□

Corollary 4.1. *Let $u(t, x)$ denote the solution to (2.6) and $\alpha \in (0, 1)$. Then for $t_0 > 0$ we may say that*

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x) \quad (4.9)$$

Proof: Follows directly from (4.7). □

We are now able to prove Lemma 3.1.

Proof of Lemma 3.1

Proof: We first define a set $\mathcal{K}(\Theta)$ as

$$\mathcal{K}(\Theta) = \{(G(1, \cdot) * y(\cdot))(x) + (F_\alpha h)(x), \quad \text{where } |y|_\rho \leq \Theta \text{ and } |h|_{L^q(0,1,L^2_\rho)} \leq \Theta\}$$

Then $\mathcal{K}(\Theta)$ is relatively compact in L^2_ρ for $\Theta > 0$. Moreover, by (4.3), (4.9) and Chebychev's inequality, and if we choose R and Θ so that $|\hat{\varphi}|_\rho < R < \Theta$ then we have

$$\begin{aligned}
\mathbf{P}\{u^{\hat{\varphi}}(1, \cdot) \notin \mathcal{K}(\Theta)\} & \leq \mathbf{P}\left[\left(\int_0^1 |Y^{\hat{\varphi}}(s, \cdot)|_\rho^q ds\right)^{1/q} > \frac{\pi\Theta}{\sin(\alpha\pi)}\right] \\
& \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Theta^q} \mathbb{E} \int_0^1 |Y^{\hat{\varphi}}(s, \cdot)|_\rho^q ds \\
& \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Theta^q} \left(1 + |\hat{\varphi}|_\rho^q\right) \\
\implies \mathbf{P}\{u^{\hat{\varphi}}(1, \cdot) \in \mathcal{K}(\Theta)\} & \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q \Theta^q} \left(1 + |\hat{\varphi}|_\rho^q\right) \\
& \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q \Theta^q} \left(1 + R^q\right)
\end{aligned}$$

Therefore if we choose $\Theta > 0$ big enough such that $\frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + R^q\right) < \epsilon$ then we get that

$$\mathbf{P}\{u^{\hat{\varphi}}(1, \cdot) \in \mathcal{K}(\Theta)\} \geq 1 - \epsilon$$

We can now deduce from this and the semigroup property of the solution to the heat equation that

$$\mathbf{P}\{u^{\hat{\varphi}}(t, \cdot) \in \mathcal{K}(\Theta)\} \geq (1 - \epsilon)\mathbf{P}\{|u^{\hat{\varphi}}(t - 1, \cdot)|_{\hat{\rho}} < R\}$$

Since we assumed that $u^{\hat{\varphi}}$ is bounded in probability in $L^2_{\hat{\rho}}$ then for and $\epsilon > 0$ we can fix $R > 0$ such that $\mathbf{P}\{|u^{\hat{\varphi}}(t, \cdot)|_{\hat{\rho}} \geq R\} < \epsilon$ and then taking $\Theta > R$ such that $\frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + R^q\right) < \epsilon$ concludes the claim. $\square$

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